Statistics & Probability

ANSWER KEY

Section 1: Central Tendency & Weighted Averages

1 A student scored $82, 88, 91,$ and 85 on four exams. What score must the student earn on the fifth exam to achieve an overall mean score of 88?

a. 90

b. 92

c. 94 ✓

d. 96

Required total for 5 exams: $5 \times 88 = 440$. Current sum: $82 + 88 + 91 + 85 = 346$. Fifth score required: $440 - 346 = 94$.

2 In a set of 7 integers: $\{4, 7, 9, x, 14, 18, 20\}$ arranged in ascending order, the median is equal to the mean. What is the value of x?

a. 12 ✓

b. 13

c. 14

d. 15

Median of 7 sorted values is the 4th value: x. Mean $= \frac{4+7+9+x+14+18+20}{7} = \frac{72+x}{7}$. Equating: $7x = 72 + x \Rightarrow 6x = 72 \Rightarrow x = 12$.

3 In a chemistry course, homework counts for 20%, labs count for 30%, and tests count for 50%. If a student earns 90 on homework, 80 on labs, and 86 on tests, what is the weighted final grade?

a. 84

b. 85 ✓

c. 86

d. 87

Weighted grade $= 0.20(90) + 0.30(80) + 0.50(86) = 18 + 24 + 43 = 85$.

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Section 2: Data Spread, Range & Standard Deviation

4 A data set consists of the numbers $\{3, 7, 8, 12, 15, 18, 21\}$. What is the interquartile range (IQR) of this data set?

a. 9

b. 11 ✓

c. 12

d. 14

The median is 12. The lower half is $\{3, 7, 8\}$ with $Q_1 = 7$. The upper half is $\{15, 18, 21\}$ with $Q_3 = 18$. Thus $\text{IQR} = Q_3 - Q_1 = 18 - 7 = 11$.

5 If every value in a data set is multiplied by 3 and then increased by 5, how is the standard deviation σ affected?

a. It increases by 5

b. **It is multiplied by** 3 ✓

c. It is multiplied by 3 and increased by 5

d. It remains unchanged

Multiplying by a constant c scales standard deviation by $|c| = 3$. Adding a constant shifts the center but does not change the spread. Hence, standard deviation is multiplied by 3.

6 Which of the following data sets has a standard deviation equal to 0?

a. $\{5, 5, 5, 5, 5\}$ ✓

b. $\{1, 2, 3, 4, 5\}$

c. $\{-2, 0, 2\}$

d. $\{10, 20, 30\}$

A standard deviation of 0 occurs if and only if all values in the data set are identical.

Section 3: Probability & Compound Events

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7 A bag contains 6 red marbles, 8 blue marbles, and 10 green marbles. If one marble is selected at random, what is the probability that it is NOT blue?

a. $\frac{1}{3}$

b. $\frac{2}{3}$ ✓

c. $\frac{1}{4}$

d. $\frac{3}{4}$

Total marbles $= 6 + 8 + 10 = 24$. Non-blue marbles $= 6 + 10 = 16$. Probability $= \frac{16}{24} = \frac{2}{3}$.

8 Two fair standard six-sided dice are rolled simultaneously. What is the probability that the sum of the two numbers rolled is greater than 9?

a. $\frac{1}{6}$ ✓

b. $\frac{5}{36}$

c. $\frac{1}{9}$

d. $\frac{7}{36}$

Outcomes with sum > 9 are sums of $10, 11, 12$: $(4,6), (5,5), (6,4), (5,6), (6,5), (6,6)$ = 6 outcomes. Probability $= \frac{6}{36} = \frac{1}{6}$.

9 Events A and B are independent. If $P(A) = 0.4$ and $P(B) = 0.5$, what is the probability that at least one of the events occurs, $P(A \cup B)$?

a. 0.70 ✓

b. 0.80

c. 0.90

d. 0.20

For independent events, $P(A \cap B) = P(A)P(B) = 0.4 \times 0.5 = 0.2$. Then $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.2 = 0.70$.

Section 4: Combinations & Permutations

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10 In how many distinct ways can a committee of 4 members be chosen from a group of 9 people?

a. 36

b. 84

c. 126 ✓

d. 3024

Selection order does not matter: $\binom{9}{4} = \frac{9 \times 8 \times 7 \times 6}{4 \times 3 \times 2 \times 1} = 126$.

11 How many different 4-digit identification codes can be formed using digits 0 through 9 if repetition of digits is not allowed and the first digit cannot be 0?

a. $4,536$ ✓

b. $5,040$

c. $9,000$

d. $3,024$

Choices for 1st digit (1-9) $= 9$. Choices for 2nd digit (0-9 excluding 1st) $= 9$. Choices for 3rd digit $= 8$. Choices for 4th digit $= 7$. Total $= 9 \times 9 \times 8 \times 7 = 4,536$.

12 A restaurant offers 4 appetizers, 6 main courses, and 3 desserts. How many distinct 3-course dinner combinations are possible?

a. 13

b. 36

c. 72 ✓

d. 144

By the fundamental counting principle: $4 \times 6 \times 3 = 72$ combinations.

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